



Economic Determinants of Passenger Railway Transport in the European Union

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Abstract This paper investigates the determinants of railway usage for passengers using traditional econometric and machine-enhanced methods. The analysis employs European Union data retrieved from The World Bank and Eurostat, spanning the years from 2000 to 2018 and covering 25 European Union countries. The size of the rail network, the gross domestic product per capita, and the fertility rate positively affect passenger traffic. Unemployment has a negative impact. Country-specific effects also appear. Policymakers, regulators, and private stakeholders in transportation, tourism, environmental protection, energy use, and economic development should focus on expanding the rail network, reducing unemployment, increasing GDP per capita, and adopting family-friendly practices. Such policies should be adapted to country-specific needs.

Keywords Rail transport · Economic growth · Railway passengers · Regularized regression

JEL Classification C51 · E60 · F47 · R41

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Introduction

This article investigates the determinants of rail transport for passengers within the European Union (EU). Using traditional and penalized regression methods, it predicts passenger traffic and explains the impact of various transport-related and macroeconomic factors. This study aims to inform the development of high-speed railway networks in the EU. Relative to other modes of transportation, railways offer benefits such as direct connections between major cities. Even failures in predictive accuracy reveal country-specific properties that may explain how drivers of rail transport affect different EU member-states.

The geographic configuration of economic activity depends heavily on the transportation of people and goods (Redding & Turner, 2015, p. 1340). Passenger railways in the EU strongly enhance the attractiveness of different regions to tourists. Conversely, visitors can stimulate railway expansion. Rail connectivity generates wider economic activity and promotes higher domestic as well as foreign investment. The economic impact of passenger rail transcends national boundaries. Rail transport creates jobs and is perceived as one of the cheapest, greenest and most convenient modes of transport.

Some of the EU's constraints on transport have arisen in combination with national guidelines, individual transport models, and technological limits (Janic, 2001). The lack of coordinated development has impeded Europe's economic integration. EU authorities have struggled to implement specific railway programs. Most investments have anticipated only the planning and design aspects of transport systems. Domestic investments usually receive the highest priority. Authorities might fail to consider similar or complementary investments in neighboring countries (European Commission, 1998).

Transportation in the EU reflects an imbalance in infrastructure and modes of transport. Individual countries and the EU as a whole often develop excessive infrastructure for one mode of transport and insufficient infrastructure for other modes. Such a development gap magnifies air pollution, noise, and congestion and reduces some of the inherent economic advantages of rail transport (European Commission, 1998).

Throughout EU history, transport has been closely linked to the establishment of the single market. Transport was essential to the fulfillment of three of the four freedoms in the 1957 Treaty of Rome: the free movement of persons, services, and goods (European Commission, 2018). EU transport policy has focused on smooth connections and networks, overcoming barriers between member-states, and creating a single European transport area. Identifying the drivers of railway passenger traffic informs EU policy-making and motivates this study.

Literature Review

This study illuminates the relationship between certain transport-related and macroeconomic variables and passenger rail traffic. This literature review identifies four themes: travel time, economic impact, environmental impact, and railway network usage.

Travel Time

Most evaluations of rail traffic assume that passengers seek to minimize travel time. Timetables reflect the actual time needed for multimodal travel (Walle & Steenberghen, 2006). Operations research can optimize rail transit schedules (including departure, arrival, and route changes) through genetic and quadratic algorithms (Wang et al., 2015). Topological network analysis can optimize transfers (Calzada-Infante et al., 2020).

Improved reliability in travel time affects the number of season ticket holders in the Dutch network (Van Loon et al., 2011). Optimization of travel times improves punctuality by 8 to 25 percent and overall welfare by 2% to 5% (Högdahl et al., 2019).

Economic Impact

Most studies found that the rollout of railway infrastructure has a positive economic impact (Alotaibi et al., 2022). Factors such as city size, industrial structure, and distance from urban cores have short-term effects on the geography of consumption and long-term effects on local growth and relocation decisions (Blanquart & Koning, 2017).

The economic impact of high-speed railways is not isolated from other political and social factors. Regulatory reform in Sweden not only reduced the cost of passenger carriage, but also transformed the entire vertical production process (Jensen & Stelling, 2007). European policymakers must consider interoperability and competition under the European Rail Traffic Management System (Laroche & Guihéry, 2013; Long et al., 2024).

Environmental Impact

As physical infrastructure, railroads are vulnerable to high temperatures and climatic fluctuations (Sanchis et al., 2020). Railway construction and expansion, in turn, can affect the environment. Although high-speed railways reduce carbon emissions, these reductions are partially offset by economic growth. Difference-in-difference methodology spots effects arising through geospatial or political change over time (Beck et al., 2010).

Railway Network Usage

Railway networks develop according to national borders (Martí-Henneberg, 2013). Regulation is tailored according to the economic, legal, and geographic peculiarities of national rail networks (Laurino et al., 2015). Broad socioeconomic factors (such as population and economic growth) affect passenger rail travel (Onderwater & Kishoon, 2017). Variables internal to railroads (capacity, ticket prices, and travel time) interact with traits of alternative modes of transport

(the cost of fuel, tolls, congestion, and parking). The purpose of travel (work, education, or leisure) also matters. Passengers who can choose among modes of transportation often minimize travel time and maximize comfort. In contrast, captive rail passengers are more sensitive to travel costs.

Data, Variables, and Methods

Data

This article surveys EU countries from 2000 to 2018 inclusive. Because Cyprus and Malta lacked sufficient data, this article covers 25 of the 27 EU countries: Austria, Belgium, Bulgaria, Croatia, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, the Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, and Sweden (Table 1).

Variables

The dataset contains 19 annual observations for 25 countries and uses eight predictors to estimate passenger traffic by rail (Table 1). Therefore, the design matrix assumes a 475×8 shape; $N=475$ for all variables and all tests. The dependent variable, *passengers*, is the number of railway passengers per capita (the ratio of passengers to the population of the country times 1 million). The independent variables are deaths per capita (*death_pc*), cars per capita (*cars_pc*), the log-transformation of the railway network (*network_log*), the fertility rate (*fertility*), the unemployment rate (*unemployment*), the inflation rate (*inflation*), GDP growth (*gdp_growth*), and GDP per capita (*gdp_pc*). Descriptive statistics are reported in Table 2.

The dataset was divided into training and test sets comprising, respectively, $1 - \frac{1}{e}$ and $\frac{1}{e}$, of all data. The training set contained 300 records and the test set contained 175. Bootstrap estimates of prediction errors reach maximum accuracy at $1 - \frac{1}{e}$ (Efron & Tibshirani, 1997), since bootstrap aggregation (bagging) will always omit $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = \frac{1}{e}$ of a dataset's observations (Breiman, 1996). Gaussian scaling proceeded exclusively with the training set (Müller & Guido, 2017, pp. 17–18). That scale was then applied to the testing set.

Each model reports the $\hat{\beta}$ vector of coefficients on the right-hand side of the regression equation as traditional, bidirectional regression outcomes (Hatcher, 2013, pp. 262–267), albeit with standardized beta coefficients as opposed to unit-specific coefficients (Siegel & Wagner, 2022, p. 387). If $\hat{y}$ represents passengers, the model may be written as:

$$\begin{aligned} \hat{y} = & \beta_0 + \beta_1 \cdot gdp_{pc} + \beta_2 \cdot gdp_{growth} + \beta_3 \cdot inflation + \beta_4 \cdot unemployment \\ & + \beta_5 \cdot fertility + \beta_6 \cdot network_{log} + \beta_7 \cdot cars_{pc} + \beta_8 \cdot deaths_{pc} + \varepsilon. \end{aligned} \quad (1)$$

Table 1 Definitions of the independent and dependent variables used in all seven models

Variables	Description	Unit	Notation
Macroeconomic variables			
GDP	Gross domestic product: Total monetary or market value of all finished goods and services produced in a country in a specific time period	Per capita purchasing power standard (PPS)	<i>gdp_pc</i>
GDP Growth	GDP growth rate compares the year-over-year (or quarterly) change in a country's economic output to measure how fast an economy is growing	Percent of GDP	<i>gdp_growth</i>
Inflation	Inflation as measured by the consumer price index reflects the annual percentage change in the cost of a basket of goods and services. The Laspeyres formula is generally used	Harmonized Index of Consumer Prices (HICP) percentage	<i>inflation</i>
Unemployment	The term unemployment refers to a situation where a person actively searches for employment but is unable to find work	Percentage of the labor force	<i>unemployment</i>
Demographic variables			
Fertility per Woman	Average number of children per woman in a female cohort subject throughout their lives to the fertility rates of a given period if they were not subject to mortality	Ratio	<i>fertility</i>
Population	A distinct group of individuals, either a nation or a group of people with a common characteristic	Number of people	Used to calculate per capita values only
Railroad-specific variables			
Passengers carried in railway	The number of passengers transported by rail multiplied by the kilometers (km) traveled	Million	(used to calculate per capita values, but not otherwise used as a predictive variable)

Table 1 (continued)

Variables	Description	Unit	Notation
Passengers per capita	The ratio of passengers carried in railway to the population of the country times 1 million	Number	<i>passengers</i> (target variable)
Railway Network	Length of the railway route in km available for train service, irrespective of the number of parallel tracks	km per 10,000 inhabitants	<i>network_log</i>
Variables concerning other modes of transportation			
Passenger cars	Passenger cars per 1000 inhabitants	per 1 000 inhabitants	<i>cars_pc</i>
Persons killed in road accidents	Passengers killed in road accidents	Number of people	<i>death_pc</i>

Data sources: Authors calculations using data from The World Bank (2022) and Eurostat (2022). The years covered are 2000 to 2018. $N=475$ for all variables

Table 2 Descriptive statistics for the independent and dependent variables used in all seven models: Central tendency, dispersion, and key quantiles

	<i>gdp_pc</i>	<i>gdp_growth</i>	<i>inflation</i>	<i>employment</i>	<i>fertility</i>	<i>network</i>	<i>cars_pc</i>	<i>deaths_pc</i>	<i>passengers</i>
mean	29,219	2.46	2.67	9.03	1.52	7974	441.27	0.00009	14,768
std	21,938	3.56	3.55	4.46	0.22	8934	110.34	0.00004	24,911
min	1621	-14.84	-4.48	1.81	1.15	274	137.00	0.00002	178
25%	13,370	1.06	1.04	5.91	1.35	2270	360.00	0.00005	1402
50%	23,099	2.52	2.10	7.86	1.47	3631	456.00	0.00008	5663
75%	42,945	4.12	3.36	10.85	1.70	10,766	512.88	0.00012	12,653
max	123,679	25.18	45.67	27.47	2.06	38,206	676.00	0.00026	110,469
count	475	475	475	475	475	475	475	475	475

Data sources: Authors calculations using data from The World Bank (2022) and Eurostat (2022). The years covered are 2000 to 2018. $N=475$ for all variables

Methods

This study applied seven linear methods: (1) ordinary least squares (OLS), (2) ridge, (3) least absolute shrinkage and selection operator (LASSO), (4) ElasticNet, (5) Bayesian Ridge, (6) automatic relevance determination (ARD), and (7) orthogonal matching pursuit. Regression serves two distinct purposes. Some applications stress the value of coefficients on the right-hand side of the regression equation. Others emphasize fitted values of the dependent variable on the left-hand side (Finkelstein, 1973, p. 1445).

This study's focus on the right-hand side as a guide to causal inference benefits from advanced regression methods that modify a baseline OLS estimate. By taming collinearity, these advanced methods sharpen causal inference through the vector of coefficients and the effect sizes that they imply. By shrinking irrelevant or redundant variables, or eliminating them outright, regularized regression can refine model specification in ways that conventional p -values cannot. Even where advanced methods do not change conclusions drawn from OLS, they supply further assurance that the model has been properly specified and that its causal inferences warrant confidence.

As will be described in greater mathematical detail, the methods depicted in Eq. (2) through Eq. (6) explicitly exact a numerical penalty against the misleadingly large coefficients (either positive or negative, and almost always in matching, offsetting pairs) that collinear, redundant variables can generate as they optimize the least-squares estimate (Mela & Kopalle, 2002). Neither the size nor even the sign of collinear coefficients reliably supports causal inference (Kim, 2019). Conscious departures from OLS through regularization can strengthen inferences by reducing collinearity (O'Brien, 2007).

Certain regularized regression methods apply the ℓ_q -norm smoothing function:

$$s(\beta) = \lambda \sum_{k=0}^K |\beta_k|^q \quad (2)$$

“for $q \geq 0$ and $\lambda \geq 0$ ” (Jakaitiene, 2019, p. 735). The key values of q are 2 and 1. The ℓ_2 norm enforces a penalty measured in Euclidean distance. The ℓ_1 norm is based on Manhattan distance. Regularized regression minimizing the ℓ_2 penalty, or ridge regression (Hoerl & Kennard, 1970), shrinks coefficients relative to OLS (James et al., 2021, p. 237).

An efficient method for eliminating inconsequential variables exploits LASSO (Tibshirani, 1996). Ridge’s ℓ_2 penalty, $\sum_1^p \beta_j^2$, gives way to LASSO’s ℓ_1 penalty, $\sum_1^p |\beta_j|$ (Hastie et al., 2009, pp. 68–69).

A hybrid of ridge and LASSO (Zou & Hastie, 2005), ElasticNet contains a regularization term combining the ℓ_2 and ℓ_1 penalties:

$$\frac{1}{2N} \left(\sum_{i=1}^N Y_i - \beta_0 - x_i^T \beta \right)^2 + \lambda \left(\frac{1-\alpha}{2} \|\beta\|_2^2 + \alpha \|\beta\|_1 \right) \quad (3)$$

where λ designates the shared ℓ_2 and ℓ_1 penalty and $\alpha \in [0, 1]$ defines each penalty’s relative share (Tay et al., 2023, p. 2). Setting $\alpha=0$ yields ridge, while $\alpha=1$ produces LASSO.

Bayesian learning frameworks also support regularization (Wipf & Nagarajan, 2007). Bayesian Ridge (MacKay, 1992) is more robust to collinearity (Shi et al., 2016). Bayesian LASSO (Casella et al., 2010) is often named for the broader family of sparse Bayesian learning methods, ARD (Tipping, 2001).

Orthogonal matching pursuit (OMP) (Rebollo-Neira & Lowe, 2002) can specify an acceptable loss in predictive accuracy. This study uses OMP to assign zero weight to variables as long as the model retains at least 95% of the r^2 value realized by OLS.

Regression methods that retain or even improve predictive accuracy relative to the OLS baseline warrant greater trust in their causal inferences. If test accuracy on holdout data approaches, meets, or beats the OLS baseline, this is evidence that supports the regularized methods’ more parsimonious expression of the $\hat{\beta}$ vector of fitted coefficients.

Empirical Results

Exploratory Data Analysis

After reporting histograms and kernel density estimates of all variables, the correlation matrix will be shown. Variable inflation factor (VIF) analysis will be conducted on the shared design matrix of the models. Country-specific effects will then be reported.

Histograms and Kernel Density Estimates

Online Supplemental Appendix (OSA) Figure 1 presents the histograms and the kernel density estimates of all predictive and target variables. These visualizations depict the variables' distributions and include statistical properties such as skewness and kurtosis.

Several variables show meaningful departures from a roughly normal distribution. *Inflation* is strongly right-skewed. *gdp_growth* is reasonably symmetrical, but has a nontrivial number of extreme observations at both ends of the distribution. To a lesser extent, *gdp_pc*, *unemployment*, and *deaths_pc* are all right-skewed.

Correlation Matrix

The correlation matrix of the variables is presented in Table 3. The positive correlation between *gdp_pc* and *cars_pc* implies that the higher the GDP per capita, the more cars that country owns. The second largest positive correlation, between *gdp_pc* and *fertility*, implies that fertility increases alongside per capita income. The positive correlation between *passengers* and *network_log* suggests that a larger network carries more passengers. Alternatively, networks may expand in response to demand.

OSA Figure 2 replicates these correlations along with the corresponding *p*-value thresholds in a heat-matrix format. The most negative correlation is between *deaths_pc* and the *gdp_pc*. As GDP increases, deaths decrease. The more money a country has, the more money it commits to transportation. A wealthier populace buys safer cars and takes more precautions. *death_pc* and *fertility_pc* are also negatively correlated. When fertility increases, deaths *per capita* decrease. This relationship may simply reflect the fact that people of childbearing age are younger. The negative correlation between *deaths_pc* and *cars_pc* might be best understood as an artifact of higher income. Though cars are more numerous, higher income ultimately reduces mortality in road accidents.

Variable Inflation Factor

Collinearity can be detected through the VIF: $VIF = \frac{1}{1-R^2}$ (Kim, 2019, p. 565–566). At $R^2 = 0$, “complete absence of multicollinearity ... minimizes the variance of the regression coefficient ...”, whereas exact multicollinearity ($R^2 = 1$) ... inflates the variance infinitely” (Kim, 2019, p. 565). $VIF > 5$ raises concerns; $VIF > 10$ “almost certainly indicates ... serious collinearity” (Menard, 1995, p. 66).

VIF analysis does not reveal obvious collinearity problems (OSA Table 1). The highest VIF value of 2.656 for *gdp_pc* falls far below thresholds for concern at 5 or 10. GDP per capita is correlated with several other variables, especially fertility and cars per capita. Positive correlation suggests that cars per capita is a surrogate for disposable income and, therefore, redundant of *gdp_pc*. GDP is also negatively

Table 3 Correlation matrix of the independent and dependent variables used in all seven models

	gdp_pc	gdp_growth	inflation	unemployment	fertility	network_log	cars_pc	deaths_pc	passengers
gdp_pc	1.000000	-0.139393	-0.266381	-0.352994	0.491927	-0.241320	0.629662	-0.537829	0.460907
gdp_growth	-0.139393	1.000000	0.144452	-0.139910	-0.137420	-0.101418	-0.242611	0.306083	-0.153780
inflation	-0.266381	0.144452	1.000000	-0.078342	-0.226374	0.013217	-0.416768	0.317617	-0.138232
unemployment	-0.352994	-0.139910	-0.078342	1.000000	-0.337545	0.068989	-0.260680	0.115063	-0.374332
fertility	0.491927	-0.137420	-0.226374	-0.337545	1.000000	-0.068470	0.231743	-0.487117	0.394633
network_log	-0.241320	-0.101418	0.013217	0.068989	-0.068470	1.000000	-0.051263	-0.133910	0.491749
cars_pc	0.629662	-0.242611	-0.416768	-0.260680	0.231743	-0.051263	1.000000	-0.402076	0.342041
deaths_pc	-0.537829	0.306083	0.317617	0.115063	-0.487117	-0.133910	-0.402076	1.000000	-0.424360
passengers	0.460907	-0.153780	-0.138232	-0.374332	0.394633	0.491749	0.342041	-0.424360	1.000000

Data sources: Author calculations using data from The World Bank (2022) and Eurostat (2022). The years covered are 2000 to 2018. $N=475$ for all variables

correlated with deaths per capita. Regularization combats collinearity of this sort in a way that OLS alone cannot.

Country-Specific Effects

Having been gathered by country, all variables reflect differences by country. All variables exhibit strong country-specific properties. OSA Table 2 summarizes each independent and dependent variable by country in the same Gaussian scale used in this paper's numerical analyses. Target data for passenger traffic appear in the final column.

Analytical Results: OLS and Regularized Regression

This section reports results for all regression methods (Table 4). Again, methods beyond OLS confirm causal inferences against the confounding effects of collinearity, in ways that conventional recitations of p -values cannot. Materially collinear variables can generate parameter estimates so extreme that their standard errors (and the p -values based upon those errors) are as misleading as the inflated coefficients to which they are attached. Consequently, the principal contribution of regularized methods takes the form of shrunken or (in the case of the ℓ_1 penalty) zeroed-out coefficients.

After introducing the ℓ_2 penalty, which shrinks model parameters without removing them from the active set, the analysis turns to methods (LASSO, ElasticNet, and ARD) under which the ℓ_1 penalty can remove entire features by assigning coefficients of zero. These methods should be compared with orthogonal matching pursuit (OMP), a machine-implemented variant of stepwise regression.

OLS

B -coefficients for traditional OLS regression appear in the first numerical column of Table 4. The OLS conventional evaluation identified four significant variables affecting passenger carriage in European rail. In declining order of effect size, they are:

- Network size, positive and significant at $p < .001$;
- Gross domestic product, positive and significant at $p < .001$;
- Unemployment, negative and significant at $p < .001$; and,
- Fertility, positive and significant at $p < .05$.

Among these variables, all significant by some conventional threshold, GDP and network size have far greater impact than unemployment or fertility. The effect size of *gdp_pc* or *network_log* is double or even triple that of *unemployment*. In contrast, *fertility* has an effect size approximately half that of unemployment. Unsurprisingly, *fertility's* significance, as indicated by conventional p -values, is weaker than

Table 4 Parameter estimates (including conventional indications of statistical significance) for all independent variables used to model rail passengers in the EU

Regression methods	OLS	Ridge	Bayesian Ridge	LASSO	ElasticNet	ARD	OMP
<i>gdp_pc</i>	0.507549***	0.486557***	0.486342***	0.472930***	0.472899***	0.495053***	0.548806***
<i>gdp_growth</i>	-0.056322	-0.055379	-0.055369	-0.035113	-0.035119	-0.023227	0.000000
<i>inflation</i>	-0.031413	-0.027664	-0.027627	-0.006881	-0.006884	0.000000	0.000000
<i>unemployment</i>	-0.206676***	-0.205559***	-0.205547***	-0.189587***	-0.189587***	-0.190758***	-0.209019***
<i>fertility</i>	0.112462*	0.115394**	0.115423**	0.103351*	0.103358*	0.102947*	0.000000
<i>network_log</i>	0.612954***	0.599905***	0.599768***	0.586788***	0.586758***	0.606197***	0.616028***
<i>cars_pc</i>	-0.021308	-0.009691	-0.009574	0.000000	0.000000	0.000000	0.000000
<i>deaths_pc</i>	0.037088	0.029684	0.029608	0.000000	0.000000	0.000000	0.000000

Statistical significance: *** $p < 0.001$; ** 0.01; * 0.05. Data sources: Authors calculations using data from The World Bank (2022) and Eurostat (2022). The years covered are 2000 to 2018. $N=475$ for all variables

the others. The remaining variables (GDP growth, inflation, the number of cars per capita, and the death rate) are statistically insignificant.

Successful application of regularized regression would confirm these causal inferences. Alternatively, if collinearity were to confound analytical results, regularization can prevent the improper attribution of causal relationships from redundant or irrelevant variables. With these observations in mind, the analysis turns to methods incorporating the ℓ_2 penalty.

ℓ_2 Regularization Methods: Ridge and Bayesian Ridge

Ridge regression implements the ℓ_2 penalty. The Bayesian variant produces similar results. Though the ℓ_2 penalty cannot eliminate coefficients, it can shrink them. Cross-validation set the ℓ_2 parameter at $\alpha=4.144285$. Both ridge and Bayesian Ridge upgraded the statistical significance of *fertility* from $p<.05$ to $p<.01$. The coefficient accompanying *fertility* is larger in either variant of ridge (0.1154) than in OLS (0.1125).

Regularized regression expands the toolkit for identifying relevant and nonredundant variables beyond null hypothesis significance testing (NHST). Marginal shifts in p -values across arbitrary conventions, such as $p<.01$ or $p<.05$, carry little analytical weight. What matters in this instance is that *fertility* retains a nonzero coefficient despite the application of regularization penalties. That nonzero coefficient, though expanded under both variants of ridge, remains smaller than coefficients on network size, GDP, and unemployment. The number of “significance stars” matters less than the effect size inferred from the absolute value of the coefficient. Regularization penalties amplify the interpretive value of effect size, relative to the conventional p -value thresholds.

This study’s use of beta coefficients reinforces the interpretive value of effect sizes. Their dimensionless nature enables beta coefficients to be directly compared with each other without regard to underlying units of measurement. The parsimonious expression of the laws of physics through “dimensionless relationships with fewer dimensionless numbers” (Xie et al., 2022, p. 1) has comparable value in economics (Kopsidas, 2018).

Advanced machine-learning methods, whether supervised or unsupervised, do not generate results directly comparable to linear parameter estimates. Regularized regression methods bridge the methodological gap between conventional linear regression and more advanced forms of machine learning. Shrunken parameter estimates (specifically, coefficients whose absolute values are closer to zero after regularization than they were under OLS) represent the primary contribution of the ℓ_2 penalty to statistical inference.

Regularization penalties do not render p -values meaningless. Rather, ℓ_2 and ℓ_1 penalties force reconsideration of causal inferences assigned under OLS alone. Shrunken parameters, paired with greater standard errors, are less likely to generate p -values that satisfy conventional thresholds such as $p<.05$ (Wasserstein et al., 2019). Indeed, the ℓ_1 penalty can remove all doubt by assigning a zero coefficient. For that reason, the focus now turns to regularization methods applying that penalty.

ℓ_1 Regularization Methods: LASSO, ElasticNet, & ARD

LASSO leads the methods that apply the ℓ_1 penalty. Even at low regularization, $\alpha=0.016019$, cars and deaths per capita exited the active set. Those coefficients having been set to zero, the ℓ_1 penalty induced sparsity. Shrinking the coefficient for *fertility* downgrades that variable's statistical significance to $p < .05$. Coefficients for *gdp_growth* and *inflation* also shrank. LASSO generates a three-feature or at most a four-feature model.

ElasticNet combines the ℓ_2 penalty of Ridge with LASSO's ℓ_1 penalty. With an ℓ_1 ratio approaching 1, this instance of ElasticNet is almost indistinguishable from LASSO. An ℓ_1 ratio of 1 produces LASSO as a special (and trivial) instance of ElasticNet. Since the shared ElasticNet penalty of $\alpha=0.016059$ is much lower than the standalone ridge penalty of $\alpha=4.144285$, the degree of ℓ_2 -based shrinkage in ElasticNet is negligible.

ARD is the Bayesian implementation of LASSO. As another method administering the ℓ_1 penalty, ARD assigns a zero coefficient to another variable: *inflation*. *gdp_growth* retains a nonzero coefficient without statistical significance. Its effect size is so small that its standard error easily traverses negative as well as positive values.

Orthogonal Matching Pursuit (OMP)

A three-feature model is not merely hypothetical. ARD results are fruitfully compared to OMP. Under OMP, a machine-implemented variant of stepwise regression, five out of eight variables could be removed with less than 5% loss in accuracy as measured by r^2 . OMP retains only the three most predictive variables: *network_log*, *gdp_pc*, and *unemployment*. *Fertility*, with the lowest effect size among variables deemed statistically significant by other methods, exits the active set.

Despite its vintage and its familiarity to economists, OMP is a disfavored method. P -values and r^2 are imperfect criteria for stepwise regression (Pace, 2008; Rose & McGuire, 2019). Regularization methods incorporating the ℓ_1 penalty (LASSO, ElasticNet, and ARD) outperform OMP in selecting features (Kumar et al., 2019; Tibshirani, 1996).

None of these methods, including OMP, explicitly rely on p -values. Instead, regularized regression relies on shrinkage to temper the effect sizes under OLS. ℓ_1 methods designate the active set in binary, all-or-nothing terms. Unlike OMP, the sparse vector of coefficients reported by LASSO or ARD includes nonzero coefficients for statistically insignificant variables. Effect sizes and p -values are best evaluated along a continuum.

Insofar as the coefficients that survive different regularization methods indicate the direction and size of each feature's effect on passenger traffic, these methods confirm this model's causal inferences. The purpose of regularization is to provide more cautious causal inference through parameter estimates in $\hat{\beta}$. The very modesty in the changes effected by regularization suggests that the design matrix was properly built. Though not every variable proved relevant, collinearity never confounded

the passenger model. Therefore, the four conventionally significant variables warrant closer attention.

Fitted Values

Although econometrics stresses “the more familiar $\hat{\beta}$ compartment” and causal inference over the “part of the toolbox marked $\hat{y}$ ” (Mullainathan & Spiess, 2017, p. 88), fitted results do convey analytical meaning. Stable, generalizable results across diverse methods, especially as applied to data not seen in training, validate causal inferences.

OSA Figure 3 presents fitted values and goodness-of-fit statistics for all methods: OLS, ridge, LASSO, ElasticNet, Bayesian Ridge, ARD, and OMP. As it must, OLS provides the most accurate predictions in training. For purposes of generalizability to new data, test-set accuracy matters more. All five regularized methods outperform OLS on test data. Ensuring the credibility of parameter estimates turns out not to sacrifice accuracy in any meaningful sense. In contrast, OMP always underperforms OLS in training and testing. OMP deliberately sacrifices accuracy as stepwise regression reduces the active set.

Modest, marginal changes in fitted values relative to OLS reflect comparably minor adjustments through regularization. Parameter shrinkage and subset selection through regularization guards against unwarranted causal inference through OLS and statistical conventions associated with NHST (McShane et al., 2019; Wasserstein et al., 2019).

Regularized regression is systematically more conservative in extracting causal inferences. Even where coefficients move within a narrow band and ℓ_1 methods induce little or no sparsity, the very modesty of those adjustments reinforces confidence in the design matrix. The $\hat{\beta}$ vector of coefficients, though not necessarily significant as a matter of statistical convention, provides credible inferences that withstand tests of collinearity and remain stable and generalizable even when applied to data not seen in training. Fitted values generated by advanced regression methods, insofar as they approach or exceed test-set accuracy under OLS, reinforce the stability of these results.

Railway carriage of passengers in the EU appears to be primarily a positive function of rail network size and GDP per capita. To a lesser extent, unemployment exerts a negative effect. Fertility appears to have a very mildly positive relationship. The convergence of solutions on both sides of the regression equation (fitted values to the left and parameter estimates to the right) strengthens these conclusions.

Discussion: Interpretation of results and policy recommendations

Three aspects of the results warrant discussion. First, nonzero parameter estimates merit closer examination as sources of causal inference and policy guidance. Second, heteroscedasticity can shed light on a model’s limitations and improve future

research. Third, departures from predictive accuracy invite a country-specific look at empirical outcomes. Geopolitically and socioeconomically diverse data from 25 EU countries predict passenger traffic in certain countries well and others poorly.

This section addresses these phenomena in turn: (1) The etiology of passenger carriage by rail; (2) successes and failures in the teleology of the models' predictions, and (3) the geodesy implied by geospatial or geopolitical divisions within the data.

Causal Inferences in $\hat{\beta}$

The most influential factor affecting passenger traffic is the size of the rail network. The larger the network, the larger the passenger volume. This is a canonical illustration of network effects (Furuga & Madanat, 2013; Na et al., 2013; Zhang & Levinson, 2007). A larger network connects more destinations, attracting proportionally more passengers. A railway network becomes more valuable with each new node and edge.

Higher GDP per capita also implies more passenger traffic. Countries with higher GDP per capita have higher levels of passenger carriage by rail. Higher economic productivity, (disposable) income, and accumulated wealth also imply the ability to expand the railway network. Although GDP per capita and network size are correlated, their low variance inflation factor and the application of regularized regression should mitigate concerns that the collinearity might confound causal inferences drawn from each predictor.

Whether a country wishes to expand its rail network depends on the costs and benefits of passenger travel by rail. Policymakers must weigh the relative impact of rail travel versus individual automobiles and urban development around rail corridors.

One of this study's striking findings is the nearly negligible impact of private car ownership. Cars and trains compete as modes of surface transportation. What cars lose in speed to high-speed rail, they presumably offset through convenience and control. However, OLS regression assigns a barely perceptible coefficient of -0.021308 . Even more tellingly, every method capable of inducing sparsity, whether through the ℓ_1 penalty (LASSO, ElasticNet, ARD) or through machine-guided stepwise regression (OMP), assigns a zero coefficient to *cars_pc*. The complete, mathematically cogent elimination of a hypothesized predictor, without resort to NHST, may be the strongest justification for regularized regression.

Although the effect is weaker than the effects arising from network size and GDP, higher birth rates are associated with higher passenger rail traffic. Over the long run, and holding constant the effects of emigration as well as immigration and the impact of internal migration between urban and rural areas, countries with higher reproductive rates will presumably experience a growth in the number and the density of urban population centers. These are the population patterns that support greater demand for passenger travel by rail.

Finally, countries with higher unemployment experience less passenger travel by train. High unemployment presumably deprives citizens of the disposable income

to travel by any modality, even the cheaper rail option. The interaction of reduced employment and income with other causal factors cannot be ruled out. Reductions in income associated with unemployment not only directly reduce the marginal propensity to travel, by train or otherwise, but also lower national economic productivity and the pool of resources by which the government and private operators can expand the railway network.

Residuals

Residuals were then scrutinized from the training of the OLS model. Breusch and Pagan (1979) reported violations of homoscedasticity: The p -values for the Lagrange multiplier and f -statistic tests of heteroscedasticity are tiny. Heteroscedasticity is prevalent throughout the behavioral and social sciences (Rosopa et al., 2013). Heteroscedasticity issues will be covered in the conclusions, and future research section.

Country-specific effects

Tables of residuals highlight country-specific effects. Their root mean squared error (RMSE) by country provides the most readily interpreted measurement of predictive error.

OSA Table 3 reports residuals for OLS and LASSO. Parameter estimates and residuals are based on combined results for the training and test sets. For purposes of unveiling country-specific effects, differences in predictive accuracy between country-specific subsets of data matter most.

The first column in OSA Table 3 reports the mean of *passengers_pc* by country. It is helpful to understand the country-specific mean in terms of Gaussian distance from the European mean. The mean among country-specific means is -0.007957 ; the standard deviation is 1.006192 . Those are modest departures from zero mean and unit variance.

The next six columns summarize, respectively, OLS and LASSO residuals for each country. Predictions are the mean fitted values for *passengers_pc* by country. The residual for each method reports the difference between the mean of ground truth (information known to be real or true) y and the mean of the fitted value $\hat{y}$. The RMSE offers a principled indicator of accuracy. The RMSE is typically a little larger than the absolute value of the mean value of residuals. Although OLS cannot be beaten in training, LASSO does outperform OLS with respect to test data. Some countries improve under LASSO, whereas the residuals for other countries.

Country-specific differences are stark. OSA Table 4 shows the five countries whose RMSE under LASSO exceeds the European mean of 0.523067 by at least one multiple of the standard deviation of 0.277099 . The most accurate value in this group, 0.812727 for Hungary, considerably exceeds even the test-set RMSE for the LASSO model, which is 0.592497 . OSA Figure 4 depicts these five outlier countries among fitted values for LASSO.

Erroneous predictions by $\pm 1z$ occur on either side of the population mean for *passengers_pc*. The model overestimates passenger travel in the relatively low-traffic countries of Romania and Ireland. Aside from geographic isolation, these countries have little in common. One is left to speculate why the model overpredicts Romanian passenger traffic, even as it delivers accurate predictions for Romania's Balkan neighbor, Bulgaria. Indeed, with respect to the most predictive variable, *network_log*, Ireland (-0.917724) and Romania ($+0.755298$) occupy opposite ends of the European distribution (OSA Table 2).

In contrast, Hungary, Austria, and France are high-traffic countries for which the model underpredicts passenger volume. The average value of *passengers_pc* for France exceeds that of Germany by nearly $1.42z$. However, a mild overestimate for German data, paired with an underestimate for France exceeding $1z$, collapses the estimates for those countries.

Entity-specific effects can help detect omitted variable bias (Agiropoulos et al., 2022, p. 2771). The regularized regression methods applied throughout this article directly remedy the opposite problem of included variable bias, or Type I error arising from the mistaken inclusion of extraneous variables (Kalnins, 2018). Extraneous variables are nuisance variables that increase random variance relative to an otherwise “reasonably designed experiment” (Jørgensen et al., 2004, p. 452). Ridge, LASSO, and their cousins address those problems.

Regularized regression indirectly addresses omitted variable bias through the residuals between the ground truth and fitted values. The examination of country-specific residuals reveals effects unique to countries within panel data in a distinct way. Whereas fixed entity effects are tested directly during regression, through the inclusion of a subarray of dummy variables within the design matrix, this approach operates *ex post*. Only after the regression is conducted are predictive failures scrutinized. These models are not only useful when they succeed. They also supply useful insights when they fail.

Policy Recommendations

This study informs policymakers, regulators, and private stakeholders in transportation, tourism, environmental protection, energy use, and economic growth. The most important driver in rail passenger traffic is the size of the rail network. Individual EU member-states and the EU should expand the railway network, especially where it is underdeveloped. Immediate benefits are reduced travel time, increased safety, punctuality, and passenger comfort. Longer-term advantages include reduced CO₂ emissions.

Unemployment has a negative impact on passenger rail traffic. Increased use of rail transport gives officials one more reason to fight unemployment. Low unemployment boosts the economy of individual countries and of the EU as a whole. This study supports policies that favor job creation and high employment as a means to boost railway usage.

GDP per capita raises passenger traffic. Aiming for higher productivity can raise railway usage by passengers. If member-states wish to promote railway transport,

they may consider subsidizing it, either directly or indirectly. Indirect subsidies may include tax benefits or more aggressive enforcement of environmental measures against cars.

Although the effect is less certain and imperfectly understood, fertility appears to favor passenger rail travel. Policies encouraging greater fertility may promote railway usage. Train travel could be made more family-friendly than competing modes of transport. Incentives include reduced-price tickets for children and improved services for families.

Regularized regression, on its own, does not identify omitted variable bias or country-specific effects. However, residuals from that analysis can accomplish both of those tasks. The presence of country-specific effects raises the possibility of segmented policies alongside uniform approaches to rail traffic. Proportional EU policymaking means that similar countries should be considered in tandem.

Unsupervised machine learning can guide further work on differences among groupings of EU countries. Deeper analysis, emphasizing country- or cohort-specific differences, might clarify the remaining inconsistencies in this analysis. For example, the failure of OMP to include fertility in its three-factor model of rail passenger traffic casts modest doubt on policy recommendations involving family-friendly rail travel. Clustering or anomaly detection may confirm or neutralize those doubts.

Conclusions and Future Research Venues

This paper applies diverse regression methods to the determinants of passenger transport by rail in the European Union. The rail network and the GDP per capita strongly and positively influence passenger rail travel. To a lesser but still statistically significant extent, the fertility rate also exerts positive influence. In contrast, unemployment has a negative impact. This variable carries less weight than network size or GDP, but more than fertility. Low VIFs and the use of regularization mitigate concerns that these conclusions may be confounded by collinearity. This is the ultimate benefit of deploying ridge, LASSO, and other regression methods alongside OLS.

Analysis of the residuals suggests country-specific variation. Such variation may interact with income and wealth effects. The worst predictions for passengers affect a diverse set of countries toward both extremes of the European distribution. Predictive accuracy, particularly if stable as applied to new data not seen during training, also reinforces the reliability of causal inferences extracted in this paper.

This study exposed entity-specific effects across the EU. Those effects invite further quantitative and qualitative research. Omitted variables latent in these country-specific effects may help explain regional patterns. The geospatial isolation of certain countries from the rest of the EU is almost assuredly relevant. The shared history of the Baltic states since 1918 may also play a role.

The idea of geodesy (literally: division of the earth) invites unsupervised machine learning. Clustering as a form of data preprocessing enables cluster-specific regressions and parameter estimates unique to each distinct cohort of countries.

Especially if combined with unsupervised machine-learning methods (such as clustering or nearest-neighbor analysis), contradictory entity effects can extend this study's work with regularized regression and find a wholly new set of applications for those methods. Effects unique to individual countries or cohorts of similar countries may tilt key predictive variables in ways that make the resulting parameter more reflective of the imbalance of sampled instances and not of any truly causal effect across the entire dataset. More aggressive regularization might expose the constellation of "significance stars" (derived from illusory p -values) as artifacts of excessive collinearity or fixed effects.

Interaction effects among hypothesized predictors seem evident. Especially because the number of proposed predictors is modest, a variant of polynomial regression can test all two-variable combinations of predictors. The sparsity-inducing properties of ℓ_1 methods should prove especially useful for eliminating most interaction effects and focusing attention on the relationships that appear both relevant and nonredundant.

Preliminary work suggests that this dataset should be amenable to machine-learning methods such as random forests. Supervised machine learning is resistant to collinearity and heteroscedasticity. Decision-tree ensembles and support vector machines alleviate excessive covariance among predictors and inconsistent variance within each data series. Problems addressed by regularized regression are distinct from heteroscedasticity. Supervised machine learning promises an effective way to address heteroscedastic data.

This paper can inform policymakers, regulators, and private stakeholders. Its results suggest that the EU should expand its rail network, especially in member-states with less extensive coverage. European authorities also need to fight unemployment and increase GDP per capita. These recommendations promote broader economic growth alongside the rail network. In addition, family-friendly practices may invite more passenger traffic.

Any changes in policy should be proportionally applied. The proportionality principle animates many areas of EU policymaking. Segmented policies that augment uniform aspects of EU policy would focus on the particularities of each country. The most underdeveloped countries may demand additional attention.

Limitations on this research include its focus on the macroeconomic, demographic, and industry-specific drivers of rail transport. A much more comprehensive model might conceive of rail transport as an equilibrium among systems of equations balancing both supply-side and demand-side factors. In that case, the appropriate estimation technique might include consumer- or household-level variables, such as household income and ticket prices. This study did not consider those variables. Expansion of the design matrix and application of advanced machine-learning methods might approach the sophistication of macroeconomic work using computable or dynamic stochastic general equilibrium.

In future research, this article's work with generalized linear models should remain useful and informative. Nonlinear machine-learning methods are harder to interpret on the right-hand side. Parameter estimates from this study would enable the proper interpretation of supervised machine learning. Those models,

despite greater accuracy, are typically more awkward and harder to interpret as guides to causal inference.

Data Availability All data come from public sources.

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